

Kai Wu

Postdoctoral Researcher in Computational Astrophysics

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RESEARCH PROFILE

Computational astrophysicist specializing in collisional stellar dynamics and direct N-body simulations of dense stellar systems. Research spans massive star clusters, compact-object populations, Population III clusters, and the dynamical evolution of planetary systems in cluster environments. Developer and code administrator of NBODY6++GPU, with experience in million-body simulations, heterogeneous GPU supercomputing, and large-scale simulation-data analysis.

ACADEMIC APPOINTMENTS

Postdoctoral Researcher

2024–present

Astronomisches Rechen-Institut (ARI-ZAH), Heidelberg University, Germany

- Lead and analyse direct N-body simulations of globular, nuclear, and Population III star clusters within the DRAGON-III programme.
- Develop coupled methods for planetary-system evolution in dense stellar environments using NBODY6++GPU and REBOUND.
- Maintain scientific software and simulation workflows across heterogeneous CPU/GPU supercomputing systems.

EDUCATION

PhD in Mathematical Sciences

2019–2024

University of Liverpool, UK

- *Thesis:* Dynamical stability of debris discs with planets in star clusters
- Supervisor: Prof. M. B. N. Kouwenhoven; research conducted off-site at Xi'an Jiaotong-Liverpool University (XJTLU).
- Full scholarship covering tuition and stipend.

BSc in Physics

2015–2019

Nankai University, China

- *Thesis:* Dynamical evolution of young open clusters in the solar neighbourhood
- Supervisor: Prof. Fang Bo; cumulative GPA: 85.3/100.
- Graduation thesis awarded Excellent Thesis (top 3%).

RESEARCH AND SOFTWARE CONTRIBUTIONS

DRAGON-III and dense stellar systems: million-body simulations of globular and nuclear star clusters over cosmic time; rotating and extremely massive Population III clusters; compact objects and multi-messenger observables.

Planetary systems in clusters: dynamical stability of debris discs and planets, survival of planetary architectures along stellar orbits, and long-term evolution of free-floating minor bodies.

NBODY6++GPU developer and lead maintainer: code developer since 2022; introduced Git-based development and CI testing; manage pull requests, issues, documentation, and user support.

Numerical-method development: implemented Fortran routines for planetary-system evolution and C/C++ interfaces for high-performance management of REBOUND simulations.

HPC and data: large-scale production management, failure diagnosis, restart validation, and reproducibility of long-running MPI/GPU simulations. Simulation and benchmarking experience on JUPITER, JUWELS, LUMI, Dongfang, Silkroad, and Kepler; CUDA/HIP platforms; analysis of more than 100 TB of DRAGON simulation data.

Numerical verification and simulation reliability: develop independent reference calculations, finite-difference and convergence tests, end-to-end validation workflows, and guarded production checks for direct N-body simulations and Galactic-potential implementations.

PUBLICATIONS

Peer-reviewed journal articles

- [1] **Kai Wu**, Ataru Tanikawa, Francesco Flammini Dotti, Marcelo C. Vergara, Boyuan Liu, Albrecht W. H. Kamlah, Manuel Arca Sedda, Nadine Neumayer, and Rainer Spurzem. *Direct N-body simulations of rotating and extremely massive Population III star clusters*. *Astronomy & Astrophysics*, 709, A142 (2026). doi:10.1051/0004-6361/202558659.

- [2] Francesco Flammini Dotti, M. B. N. Kouwenhoven, **Kai Wu**, Abbas Askar, Peter Berczik, Mirek Giersz, Rainer Spurzem, and Ian Dobbs-Dixon. *Dynamical evolution of massless particles in star clusters with NBODY6++GPU-MASSLESS: II. The long-term evolution of free-floating comets*. Astronomy & Astrophysics, 706, A219 (2026). doi:10.1051/0004-6361/202557334.
- [3] Guimei Liu, Yu Zhang, Jing Zhong, Li Chen, Xiangcun Meng, and **Kai Wu**. *Formation and evolution of new primordial open cluster groups: Feedback-driven star formation*. Astronomy & Astrophysics, 696, A117 (2025). doi:10.1051/0004-6361/202452774.
- [4] Songmei Qin, Jing Zhong, Tong Tang, Yueyue Jiang, Long Wang, **Kai Wu**, Friedrich Anders, Lola Balaguer-Núñez, Guimei Liu, Chunyan Li, Jinliang Hou, and Li Chen. *Unveiling the binary nature of NGC 2323*. Astronomy & Astrophysics, 693, A317 (2025). doi:10.1051/0004-6361/202452962.
- [5] **Kai Wu**, M. B. N. Kouwenhoven, Francesco Flammini Dotti, and Rainer Spurzem. *Influence of planets on debris discs in star clusters - II. The impact of stellar density*. Monthly Notices of the Royal Astronomical Society, 533(4), 4485–4499 (2024). doi:10.1093/mnras/stae2067.
- [6] **Kai Wu**, M. B. N. Kouwenhoven, Rainer Spurzem, and Xiaoying Pang. *Influence of planets on debris discs in star clusters - I. The 50 au Jupiter*. Monthly Notices of the Royal Astronomical Society, 523(4), 4801–4817 (2023). doi:10.1093/mnras/stad1673.

Submitted manuscripts

- [7] Philip Cho, **Kai Wu**, Francesco Flammini Dotti, Taras Panamarev, Shuo Li, Shiyan Zhong, Klaus Reuter, and Rainer Spurzem. *Simulating tidal disruption events in nuclear star clusters*. Submitted to Astronomy & Astrophysics, arXiv:2609.31083 (2026).

Accepted conference proceedings

- [8] M. B. N. Kouwenhoven et al. (including **Kai Wu**). *Planetary systems in dense stellar environments*. Journal of Physics: Conference Series, 1523, 012011 (2020). doi:10.1088/1742-6596/1523/1/012011.
- [9] **Kai Wu**, Philip Cho, Rainer Spurzem, Long Wang, Francesco Flammini Dotti, and Vahid Amiri. *DRAGON-III simulation: modelling million-body globular and nuclear star clusters*. Proceedings of IAU Symposium 398, arXiv:2510.03933 (2025).
- [10] Philip Cho, **Kai Wu**, Francesco Flammini Dotti, Taras Panamarev, and Rainer Spurzem. *Modeling Tidal Disruption Events and Compact Object Plunges in Nuclear Star Clusters*. Proceedings of IAU Symposium 398, arXiv:2602.08574 (2026).
- [11] Yuzhe Song, Debatri Chattopadhyay, Jarrod Hurley, Rainer Spurzem, Francesco Flammini Dotti, and **Kai Wu**. *Modelling Millisecond Pulsar Populations in Globular Clusters with NBODY6++GPU*. Proceedings of IAU Symposium 398, arXiv:2511.08850 (2025).

Manuscripts in preparation

- [12] **Kai Wu**, Francesco Flammini Dotti, Marcelo C. Vergara, Vahid Amiri, and Rainer Spurzem. *DRAGON-III simulation - I. Multi-messenger observations in the computer and comparison with observed astrophysical objects*. In preparation.
- [13] **Kai Wu**, M. B. N. Kouwenhoven, Rainer Spurzem, and Francesco Flammini Dotti. *Linking the survivability of planets in star clusters with the orbits of their host stars*. In preparation.

HONOURS AND AWARDS

\$360 equivalent, ICISE, accommodation support for conference Modelling and Observing Dense Star Clusters.	2026
Selected participant (40 places), <u>Astro-CC Pilot Virtual Observatory school</u> , Madrid. Travel and accommodation covered by the organisers.	2025
€1,500, Young Researcher and Innovator Conference Grant, <u>COST Action CA22133</u> . Awarded to present at the European Astronomical Society Annual Meeting.	2025
€450, Travel Grant, Astronomische Gesellschaft. Support to present two talks at the Annual Meeting of AG.	2025
Excellent Teaching Assistant Award (top 2%), XJTLU	2023
Best Oral Presentation Award (top 5%), XJTLU Postgraduate Research Symposium	2022
Best Poster Award (top 5%), XJTLU Postgraduate Research Symposium	2021
Full-scholarship PhD, tuition waiver and stipend, XJTLU	2019–2023
Excellent Graduation Thesis (top 3%), Nankai University	2019
Boling Scholarship (top 10%, three times) and Second-class Scholarship (top 8%), Nankai University	2015–2017
Second Prize (top 10%), Nankai Physicists' Tournament	2016

INVITED COURSES AND SEMINARS

Courses and tutorials

Shanghai Astronomical Observatory: 30-hour doctoral course on N-body simulations with <u>NBODY6++GPU</u>	Aug. 2023
Spurzem's International Seminar: four-hour tutorial on Git and GitHub for NBODY6++GPU development and collaboration	Apr. 2022

Seminars

Max Planck Institute for Astrophysics, Garching (colloquium): Multi-scale Stellar Dynamics in Dense Clusters: From Planetary Debris to Black Hole Seeds	Apr. 2026
ARI-ZAH, Heidelberg University: From Debris Disks to Million-Body Clusters: Planetary Perturbations, Stellar Interactions, and DRAGON-III	Jul. 2025
Zhejiang University: Dynamics of debris discs with planets in star clusters	Oct. 2024
KIAA-Peking University: Dynamics of debris discs in star clusters	Jul. 2023
Shanghai Astronomical Observatory: An in-depth introduction to numerical star-cluster simulations	Jul. 2023

SELECTED CONFERENCE PRESENTATIONS

Annual Meeting of the German Astronomical Society (AG2026), Garching – From million-body star clusters over cosmic time to black-hole growth in primordial systems	Sep. 2026
Bridging It: Connecting Observations and Models of (Exo)planets and their Formation, MPIA, Heidelberg – SCULPT: star-cluster and planetary-system dynamics as physically motivated prescriptions for planet population synthesis	Sep. 2026
MODEST-26: Modelling and Observing Dense Star Clusters, Quy Nhon – Rotation-driven collapse and black-hole growth in primordial star clusters	Aug. 2026
14th <u>Silk Road</u> Meeting, Tashkent – From million-body star clusters over cosmic time to black-hole growth in primordial systems	May 2026
Annual Meeting of the German Astronomical Society (AG2025), Goerlitz – two contributed talks on DRAGON-III and planetary systems in star clusters	Sep. 2025
European Astronomical Society Annual Meeting, Cork – Dynamical stability of debris discs with planets in star clusters	Jun. 2025
IAU Symposium 398 / MODEST-25, Seoul – DRAGON-III million-body globular and nuclear star clusters	Jun. 2025
Gas Accretion in Planet Formation, MPIA – Star clusters, protoplanetary discs, and debris-disc evolution	Mar. 2025
SPP 1992 Final Colloquium, Berlin – Dynamical evolution of planetary systems in star clusters	Mar. 2024

TEACHING AND ACADEMIC TRAINING

Associate Fellow of the Higher Education Academy (AFHEA), Advance HE	2024
Teaching qualification aligned with the UK Professional Standards Framework.	
Part-time Lecturer – IB Calculus, Suzhou Research Institute of Xi'an Jiaotong University	Jan.–Jul. 2022
Designed and delivered 105 hours of calculus instruction for an international high-school preparatory programme.	
Teaching Assistant / Tutor, Xi'an Jiaotong-Liverpool University	2019–2023
- Delivered approximately 30 hours of English-language teaching in physics experiments, calculus, advanced linear algebra, and statistical methods. - Received the School of Mathematics and Physics Excellent Teaching Assistant Award (top 2%).	
Invited Instructor – Doctoral N-body Course, Shanghai Astronomical Observatory	Aug. 2023
Delivered a 30-hour practical course on simulating star clusters with <u>NBODY6++GPU</u>.	
Invited Tutorial Instructor, Spurzem's International Seminar	Apr. 2022
Delivered a four-hour tutorial on Git, GitHub, and collaborative scientific-software development.	

ACADEMIC SERVICE AND MEMBERSHIPS

Referee, The Astrophysical Journal	2026–present
Referee, Astronomy & Astrophysics	2026–present
Referee, Planetary and Space Science	2025–present
Member, <u>European Cooperation in Science and Technology</u> Action <u>CA22133</u> “The Birth of Solar Systems”	2023–present
Member, <u>Silk Road Collaboration</u> for stellar dynamics and computational astrophysics	2019–present

SELECTED ACADEMIC VISITS

National Astronomical Observatories of China and KIAA-Peking University – visitor of Dr. Shuo Li	Dec. 2024; Jun. 2025
Frankfurt Institute for Advanced Studies – visitor of Prof. Volker Lindenstruth	Jan. 2025
Heidelberg University – guest researcher funded by DFG SPP 1992	Mar.–Apr. 2024
National Astronomical Observatories of China – visiting scholar funded by XJTLU	Dec. 2019
CITA, University of Toronto – visiting student of Prof. Yanqin Wu	Jul.–Oct. 2018
Space Telescope Science Institute / Johns Hopkins University – visiting student of Prof. Christine Chen	Jun.–Sep. 2017
Earlier visits: Purple Mountain Observatory; National University of Singapore; Nanyang Technological University	2016

TECHNICAL EXPERTISE

Scientific programming: Python, Fortran, C/C++, Bash; NumPy, SciPy, pandas, multiprocessing
Parallel and GPU computing: MPI, OpenMP, CUDA, HIP; heterogeneous CPU/GPU systems
AI-assisted research: tool-enabled LLM coding/research agents for scientific software, numerical verification, HPC debugging, and research automation, with test- and physics-based validation. Multi-agent orchestration.
Scientific software: NBODY6++GPU, PeTar, REBOUND, MERCURY
Data and visualisation: HDF5, Matplotlib, Plotly; large-scale simulation-data processing
Development workflows: Git, GitHub, CI/CD, Docker, documentation, issue and release management
Typesetting and reproducibility: LaTeX, Linux-based research workflows